

Beyond violence: The educational consequences of insurgent rule

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Abstract

This paper studies the effects of rebel governance on schooling, distinguishing them from the impacts of violence. Using individual-level panel data from northeast Nigeria (2009–2023) and a difference-in-differences design, I compare children in areas temporarily occupied by Boko Haram to those in nearby areas exposed to violence but not occupation. Rebel governance—not violence—drives large and persistent educational losses. Children in occupied areas gain roughly three-quarters of a year less schooling per year of exposure; six years later, they have completed 1.5 fewer years, with no recovery. The effects are gendered: girls experience larger short-term attendance declines, boys greater long-term attainment losses. Standard explanations—infrastructure damage, health shocks, child labor—find limited support. Instead, the evidence points to a disruption of schooling norms under rebel rule, consistent with Boko Haram’s anti-education governance and the persistence of attitudinal shifts after the restoration of state authority.

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1 Introduction

When insurgents shift from intermittent attacks to territorial occupation, the nature of disruption to civilian life changes fundamentally. Violence undermines education through destruction and fear, but occupation introduces structures of governance: public goods such as education fall under the authority of the occupying group, which may use them to suppress, influence, or coerce the local population (Berman 2003; Mampilly 2021; Kalyvas 2006; Maynard 2019; Yakter and Harsgor 2022). This paper examines how living under insurgent governance affects educational outcomes, showing that it is rebel rule—not violence per se—that drives large and persistent losses in schooling.

Using individual-level panel data on school-aged children in northeast Nigeria from 2009 to 2023, I compare children who experienced Boko Haram’s territorial occupation to children in neighboring areas exposed to violence and conflict risk but not to insurgent rule. A difference-in-differences design identifies the effects of occupation on the annual progression of years of schooling (YoS)—the primary outcome—and on cumulative educational attainment.

The central finding is that rebel governance has effects on education that are an order of magnitude larger than those of violence alone. Children in occupied areas gain roughly three-quarters of a year less schooling per year of exposure, meaning that the occupation largely eliminated educational progression for affected children. This disruption does not dissipate: six years after the occupation ended, children in previously occupied areas have completed 1.5 fewer years of schooling—equivalent to roughly 30 percent of the regional adult average—with no evidence of recovery. School attendance also falls sharply, by approximately 15 percentage points. Violence exposure, measured through counts of Boko Haram-related events near the household, has no statistically significant effect on either outcome. The effects are strongly gendered: girls experience a far steeper short-term decline in attendance, while boys suffer larger and more persistent losses in cumulative years of schooling.

The evidence on mechanisms is most consistent with a disruption of schooling norms under rebel governance. The interaction between occupation and Muslim household status amplifies the attendance effect; the gendered pattern mirrors Boko Haram’s specific targeting of girls’ education; and the persistence of effects years after the occupation ended points to a durable shift in attitudes rather than a temporary disruption of access. Standard conflict-related explanations—child labor, early marriage, health shocks, and infrastructure destruction—find limited support, though imprecision in the infrastructure estimates means that moderate effects cannot be ruled out.

This paper contributes to a growing literature on rebel governance (Arjona 2016, 2017; Mampilly 2021; Stewart 2018; Rubin 2020), to work on Islamic insurgents and education (Bertoni et al. 2019; Ajogbeje and Sylwester 2024; Peters 2014; Afzal 2020), and to the broader study of conflict and human capital (Justino 2011; Swee 2015; Shemyakina 2011; Di Maio and Nisticò 2019; Brown et al. 2019; Guerra-Cújar et al. 2024; Bove et al. 2024). The most closely related study, Bertoni et al. (2019), examines the effect of Boko Haram-related *violence* on schooling using the same survey

data.¹ The central distinction lies in the treatment definition: violence exposure versus territorial occupation. A nesting exercise confirms that the occupation effect remains large and significant after controlling for violence, indicating that the results capture something distinct from and additional to violence exposure (Appendix Table D.2).

The paper proceeds as follows. Section 2 provides background on Boko Haram and education in northeastern Nigeria. Section 3 describes the data and sample. Section 4 outlines the empirical strategy. Results are presented in Section 5, mechanisms in Section 6, and Section 7 concludes.

2 Background

2.1 Insurgency

Although specific ideologies vary, many organizations that self-identify as Islamic insurgent groups share several core positions: the use of violence against those perceived as opposing Islamic principles, the aim of establishing an Islamic state, and the rejection of secular governance (Crisis group 2016). This rejection frequently extends to secular (“Western”) education. When these groups occupy territory, they tend to impose restrictions on education, frequently denying access to formal schooling and engaging in violent attacks on educational institutions (GCPEA 2020).²

Boko Haram exhibits all of these characteristics, combining a rejection of democracy with a strongly anti-education ideology. As one of the largest militant groups in Africa, it has been primarily active in the Lake Chad Basin, particularly in northeastern Nigeria. The group launched its armed insurgency in 2009. In spring 2014, Boko Haram began occupying territory in northeastern Nigeria, formally declaring a caliphate in August that year. The group maintained control over these areas until early 2015, when Nigerian military operations reclaimed the territory (see Figure I.5).

According to interviews with former residents conducted by Amnesty International (2015), Boko Haram imposed a system of rule grounded in its anti-education ideology. The prohibition of secular schooling was central to its governance (Center for International Security and Cooperation 2018; Omenma et al. 2020; Anugwom 2018; Thurston 2016; CFR 2018). A more detailed account of Boko Haram’s ideology and the sequence of events leading to its territorial control is provided in Appendix A.

2.2 Education in Northeastern Nigeria

The 2004 Compulsory, Free Universal Basic Education Act mandates nine years of free primary and junior secondary education. In practice, however, Nigeria has the highest number of out-of-school

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1. Bertoni et al. (2019) rely on the same NGHS question—“Are you currently attending school?” or “Did you attend school in the previous school year?”—which they refer to as enrollment. I use the term attendance to remain consistent with the survey wording.
 2. Examples include Ansarul Islam in Burkina Faso and Mali; Jama’at Nusrat ul-Islam wal-Muslimeen (JNIM) in Mali; Islamic State of Khorasan Province (ISKP) in Afghanistan; ISIL in Iraq; the Taliban and al-Qaeda in Pakistan; al-Shabaab in Somalia; and Boko Haram in Nigeria.

children in the world and some of the lowest educational outcomes in sub-Saharan Africa (Unicef 2015; Abdullahi and Abdullah 2014). The situation is especially acute in the North East. Afzal (2020) identifies four key explanations: the education system is viewed as a colonial imposition; educational access during the colonial period was limited in the North; the system is seen as failing to deliver returns; and Western education is associated with political corruption. These perceptions contributed to local support for Boko Haram’s anti-education platform.

3 Data and Sample

3.1 Data sources

I construct a panel dataset at the individual–school-year level covering the period 2009–2024, combining five waves of the Nigerian General Household Survey (NGHS) with the COVID-era National Longitudinal Phone Survey.³ Across all waves, respondents were asked about school participation and years of completed education. I refer to the binary participation indicator as attendance to remain consistent with the survey wording and to emphasize actual participation rather than formal registration.

Additional variables used to explore potential mechanisms are drawn from the NGHS, including indicators of child labor, health, and marital status. I also use a follow-up question asked of all children not attending school—“What is the main reason you are not attending school?”—to construct a measure of disinterest in education. An important limitation is that this variable is observed only for non-attenders; I return to this issue in the mechanisms discussion.

To ensure a consistent panel, I restrict the sample to individuals born between 2001 and 2008, guaranteeing that all respondents were of mandatory school age both before and immediately after the occupation period. I further restrict the sample to rural households in LGAs covered in all survey waves and include only individuals observed at least twice—both before and after the treatment (see Figure I.2). These restrictions ensure a consistent and balanced panel across waves (Table C.1).

Data on territorial occupation come from the IOM (2015), supplemented with information on territorial seizures from the Armed Conflict Location and Event Dataset (ACLED) (Raleigh et al. 2010).⁴ I use ACLED both to validate IOM territorial control data and to construct measures of violence exposure. Due to anonymization in the NGHS and geographic uncertainty in event coordinates, I define exposure as any Boko Haram–related event within a 10 km radius of a household’s location, lagged by one year. For robustness, I vary the radii, perpetrators, and targets, and replicate all violence measures using the UCDP Georeferenced Event Dataset (Sundberg and Melander 2013). An overview of all datasets and key variables is provided in Table C.2.

3. The phone survey was launched during the COVID-19 pandemic and consists primarily of households included in the NGHS. The questionnaire closely follows that of the NGHS, particularly regarding education. I restrict the analysis to households observed in the NGHS datasets.

4. ACLED collects event-level data from media reports and a network of local journalists, informants, and NGOs. All reports are cross-referenced to verify whether specific incidents are attributable to Boko Haram.

3.2 Outcome variables

The primary outcome is the annual change in years of schooling (increase in YoS), which captures the flow of educational investment: whether a child progressed by the expected amount in a given school year. This measure directly reflects contemporaneous disruption and is not subject to the mechanical persistence that characterizes cumulative stock measures. As a complement, I examine cumulative years of schooling (YoS), capturing the total magnitude of the resulting deficit. School attendance, a binary indicator, serves as an additional outcome.

3.3 Treatment and control group

The treatment is defined as exposure to Boko Haram’s temporary territorial occupation between 2014 and 2015.⁵ Occupied areas are identified using territorial control maps from IOM (2015), cross-referenced with ACLED to determine the timing and location of territorial transfers.⁶

The control sample is restricted to individuals in LGAs that were never occupied and that directly border occupied areas (see Figure I.3), ensuring that treated and control individuals were plausibly exposed to similar perceived threats.⁷

The final sample includes 316 treated and 1,457 control observations across sixteen LGAs. As shown in Table C.3, the two groups are broadly similar at baseline. Table C.4 shows that communities in occupied LGAs are somewhat better endowed in terms of infrastructure, while control areas experienced more conflict events. There are no significant pre-existing differences in support for Boko Haram across bordering LGAs.

4 Empirical Approach

4.1 Estimation

I estimate the following difference-in-differences model:

$$Y_{i,t,j} = \alpha_i + \lambda_t + \beta(D_t \times T_{i,j}) + \sigma X_{i,j,t} + \epsilon_{i,j,t} \quad (1)$$

Here, $Y_{i,t,j}$ denotes the outcome for individual i in LGA j and school year t . D_t is a post-treatment indicator equal to 1 for school years from 2014 onward. $T_{i,j}$ equals 1 for individuals in occupied LGAs. β captures the treatment effect. α_i and λ_t denote individual and school-year fixed effects, respectively.⁸

5. Boko Haram’s occupation began in May 2014 and ended by spring 2015. As shown in Figure I.5, government forces quickly reclaimed nearly all occupied areas.

6. Two LGAs—Askira/Uba and Geidam—do not appear in ACLED as experiencing territorial transfer events. However, IOM classified both as fully occupied, and independent media reports confirm their occupation (BBC 2015; France24 2014; Al Jazeera 2014; Anadolu Agency 2015). Accordingly, they are treated as occupied.

7. As a robustness check, I extend the control group to include all LGAs in the North East region and use propensity score matching. Results are provided in the Appendix.

8. The YoS variable reflects the cumulative number of years of schooling and increases by one per year of attendance. The increase in YoS captures whether a child progressed by the expected amount in a given year.

The vector $X_{i,j,t}$ includes time-varying controls: indicators for whether the child’s father works in agriculture, gender, whether the child is of school-going age (5–15), and whether the household head is Muslim; as well as household size, child’s age, and whether a state of emergency was declared prior to the occupation.

The preferred specification excludes measures of violence exposure and perceived conflict risk, as these are themselves potentially affected by the occupation and may constitute “bad controls” (Angrist and Pischke 2009). Including them as additional controls does not materially change the estimates (Table 2; Appendix Tables D.1 and D.2).

Standard errors are clustered using the Conley (1999, 2008) HAC method, allowing for spatial correlation within 20 km and temporal correlation across three school years. Given the small number of treated clusters (four LGAs), I supplement this with wild cluster bootstrap, randomization inference, and leave-one-out exercises (Appendix F).

4.2 Validity

The NGHS excludes areas deemed unsafe for enumerators, and several LGAs were dropped starting in 2013—before the onset of occupation. Within the remaining sample, control areas were often closer to active conflict zones and experienced somewhat higher levels of violence, suggesting that any bias from differential survey coverage would likely attenuate the estimated effects.

Attrition is extremely low—less than one percent—and unrelated to treatment status (Appendix Table C.5). At baseline, treatment and control communities are highly comparable, with no significant pre-existing differences in support for Boko Haram across bordering LGAs (Table C.4).

I test the parallel trends assumption using pre-treatment event-study estimates (Figure 1). For years of schooling, the pre-treatment trajectories are closely aligned. For attendance, there is a modest but statistically insignificant divergence, so the attendance results should be interpreted with some caution. Subgroup pre-trends by gender and religion are broadly supportive of the design (Appendix Figures C.1–C.4).

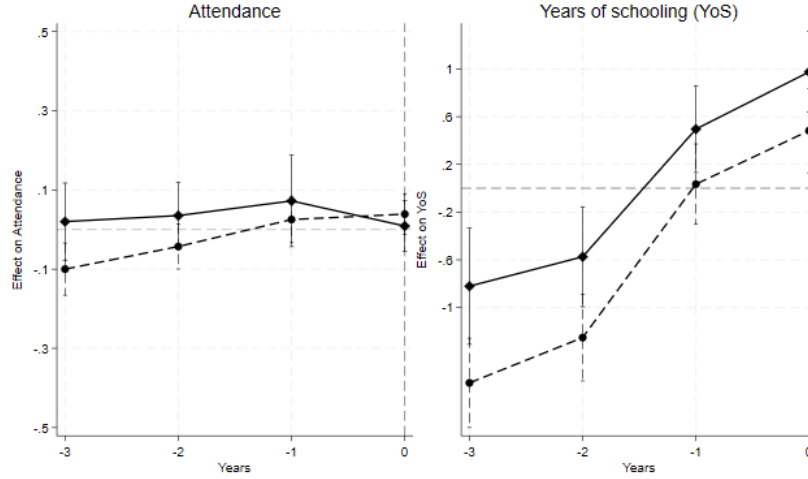
5 Results

5.1 The effect of occupation on years of schooling

Table 1 reports the preferred specification. Children in occupied areas gain roughly three-quarters of a year less schooling per year of exposure (coefficient -0.74 , $p < 0.01$), implying that the occupation largely eliminated educational progression for affected children. The cumulative consequences are substantial: children in occupied areas completed, on average, 1.48 fewer cumulative years of schooling during the post-occupation period, equivalent to more than 30 percent of average regional adult attainment. Figure D.1 illustrates a sharp decline during and immediately after occupation, followed by a persistent gap that shows no evidence of recovery.⁹

9. Figure D.2 plots attendance and YoS trends for the treatment and control groups over the full observation period. There is no visible evidence of a substantial COVID-19 effect on attendance.

Figure 1: Pre-event study: change in school attendance and YoS



Note: The graph shows the conditional differences between school attendance and YoS for the treatment (solid line) and control group (dashed line) for the pre-treatment school years. The confidence intervals are 95%.

School attendance falls by roughly 15 percentage points (25 percent) following occupation. Pre-treatment trends for attendance show a modest but statistically insignificant divergence (Figure 1), so these results should be interpreted with some caution.

Table 1: The effect of insurgent rule on education (preferred specification)

	(1)	(2)	(3)	(4)	(5)	(6)
	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Attendance	Attendance
Occupied	-0.7416*** (0.1907)	-0.7597*** (0.1902)	-1.4805*** (0.3447)	-1.3880*** (0.3433)	-0.1526** (0.0763)	-0.1634** (0.0780)
Boko Haram events	0.0134*** (0.0036)		-0.0060 (0.0079)		-0.0010 (0.0018)	
Conflict risk	-0.0095 (0.0197)		0.0253** (0.0124)		-0.0028 (0.0017)	
Mean(Y)	0.762	0.762	3.369	3.369	0.614	0.614
Observations	1,447	1,447	1,758	1,758	1,647	1,647
R-squared	0.019	0.018	0.059	0.053	0.036	0.035

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. “Occupied” captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Odd columns include BH events within 10km and conflict risk as controls; even columns exclude all violence-related controls. Additional controls: household size; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

5.2 Violence versus occupation

Violence exposure has no statistically significant effect on years of schooling or attendance. Including violence and conflict risk as additional controls barely affects the occupation estimates (Table 2). A conflict risk measure based on distance to Boko Haram’s stronghold shows a positive coefficient for years of schooling, indicating that children farther from the insurgent base have better outcomes. These patterns confirm that the occupation effect captures the consequences of insurgent rule rather than correlated violence exposure. Additional specifications—including an incremental decomposition of violence, occupation, and conflict risk (Appendix Table D.1) and a nesting exercise controlling for Bertoni-style violence measures (Appendix Table D.2)—reinforce this conclusion.

Table 2: The effect of insurgent rule on education (with violence controls)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Occupied	-0.7416*** (0.1907)	-0.6200*** (0.2070)	-0.2194 (0.1785)	-1.4805*** (0.3447)	-1.6894*** (0.3447)	-1.2122*** (0.3792)	-0.1526** (0.0763)	-0.0068 (0.0581)	-0.0291 (0.0587)
Occupation*Female		-0.2898 (0.2046)			0.4936* (0.2644)			-0.3492*** (0.1243)	
Occupation*Muslim			-0.9310*** (0.2550)			-0.4991 (0.4878)			-0.2152* (0.1245)
Boko Haram events	0.0134*** (0.0036)	0.0134*** (0.0036)	0.0132*** (0.0037)	-0.0060 (0.0079)	-0.0060 (0.0079)	-0.0061 (0.0079)	-0.0010 (0.0018)	-0.0010 (0.0018)	-0.0011 (0.0018)
Conflict risk	-0.0095 (0.0197)	-0.0091 (0.0197)	-0.0074 (0.0197)	0.0253** (0.0124)	0.0250** (0.0124)	0.0256** (0.0124)	-0.0028 (0.0017)	-0.0025 (0.0017)	-0.0026 (0.0017)
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	1,447	1,447	1,758	1,758	1,758	1,647	1,647	1,647
R-squared	0.019	0.020	0.024	0.059	0.060	0.060	0.036	0.045	0.039

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. “Occupied” captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior and a measure for conflict risk; household size; whether it is an agricultural household; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

5.3 Gender heterogeneity

Columns (5) and (8) of Table 2 reveal clear gender differences. Girls experience smaller long-run losses in cumulative years of schooling than boys (the Occupation $\times$ Female coefficient is positive), but they are 57 percent less likely to attend school immediately following occupation (the Occupation $\times$ Female coefficient on attendance is -0.35 , $p < 0.01$). The short-term effects are thus more severe for girls, while the cumulative, long-run losses in years of schooling are greater for boys.

Figures D.3 and D.4 illustrate this asymmetry. Before the occupation, girls in treated communities had slightly higher attendance than controls; their sharp post-occupation decline erased this advantage. Boys started from lower baseline levels, so even a modest disruption widened pre-existing gaps.

Robustness The results are stable across a wide range of specifications: varying violence radii, alternative conflict definitions, duration of occupation as a continuous treatment, propensity score

matching, the full North East sample, and LGA-specific linear time trends. Adding LGA trends attenuates the occupation coefficient on cumulative YoS and attendance, but the effect on annual grade progression remains large and, if anything, increases. When using UCDP as an alternative violence source, the YoS results remain robust, though the attendance effect becomes insignificant in some specifications. All results survive wild cluster bootstrap inference, randomization inference, and leave-one-out-LGA exercises. Full details are reported in the Appendix (Sections F and G).

6 Mechanisms

I examine several potential channels through which occupation may have affected schooling. No single mechanism fully accounts for the observed patterns, but the evidence is most consistent with a disruption of schooling norms under rebel governance.

6.1 Norm disruption under rebel governance

Three pieces of evidence support this interpretation. First, the interaction between occupation and Muslim household status is negative and significant for attendance (Table 2, column 9), suggesting that children in households sharing a religious identity with Boko Haram experienced a larger decline in school participation—though Muslim identity is at best a rough proxy for ideological alignment, given Boko Haram’s violence against Muslim communities that did not share its views (Thurston 2016). Second, the strongly gendered pattern of effects mirrors Boko Haram’s specific targeting of girls’ education. Third, the persistence of effects—with no evidence of recovery six years after the occupation ended—is consistent with a durable shift in attitudes rather than a temporary disruption of access.

6.2 Parental attitudes toward schooling

Among children not attending school, those in previously occupied areas are more likely to cite parental disinterest as the main reason for non-attendance (Table E.3), and this effect is disproportionately driven by girls. However, because this variable is observed only for non-attenders and conflates child and parental preferences, I interpret it as suggestive evidence consistent with the broader pattern of norm disruption rather than as definitive evidence of a specific causal channel.

6.3 Alternative mechanisms

I evaluate several other channels. Infrastructure damage is inconclusive: while the estimated coefficients on lack of schools and teachers are negative, they are imprecisely estimated and the confidence intervals are wide (Appendix Table E.5). There is no evidence that the occupation worsened child health, increased early marriage, or shifted time use toward household responsibilities or child labor (Appendix Tables E.1–E.2). Nor is there evidence of reduced returns to education: individuals with at least primary education remain more likely to be employed, and women with primary education

are 22 percentage points more likely to be employed than similarly educated men (Appendix Table E.4).

7 Concluding remarks

While the effects of conflict on civilian behavior have been widely studied, existing empirical work rarely distinguishes between exposure to violence and exposure to insurgent governance. Given that an estimated one-fourth to one-third of insurgent groups eventually establish territorial control (Wood 2008; Arjona 2016; Stewart 2018; Rubin 2020), understanding the consequences of rebel rule for children’s educational outcomes is both timely and policy-relevant.

Using individual-level panel data from northeast Nigeria, this paper has shown that Boko Haram’s temporary territorial occupation had severe and lasting effects on education. Violence alone has no significant effect on schooling; occupation reduces annual educational progression by nearly a full year. These disruptions accumulate into persistent deficits—1.5 fewer years of schooling six years later, with no evidence of recovery. The consequences are distinctly gendered, with girls experiencing larger short-term attendance declines and boys suffering greater long-term attainment losses.

The evidence on mechanisms points toward a disruption of schooling norms under rebel rule. The interaction with Muslim household status, the gendered effects, and the persistence of deficits are all consistent with a durable shift in attitudes. Standard conflict-related explanations find limited support, though imprecision in some estimates means that moderate effects of infrastructure damage cannot be fully excluded.

These findings demonstrate that non-state rule can transform the social and behavioral foundations of human capital investment in ways that persist long after formal authority is restored. The results carry direct implications for post-conflict education policy: restoring infrastructure and access alone is unlikely to reverse these deficits. Effective recovery will require addressing the attitudinal shifts that insurgent governance leaves in its wake.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process: During the preparation of this work the author used Claude code in order to write code for the analysis, and check the manuscript for consistency. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

- Abdullahi, Danjuma, and John Abdullah. 2014. “The political will and quality basic education in Nigeria.” *Journal of Power, Politics & Governance* 2 (2): 75–100.
- Afzal, Madiha. 2020. “From "Western Education is Forbidden" to the World’s Deadliest Terrorist Group: Education and Boko Haram in Nigeria.”

- Ajogbeje, Korede, and Kevin Sylwester. 2024. "How conflict affects education: Differences between Boko Haram and Farmer-Herder conflicts in Nigeria." *World Development* 177:106540.
- Al Jazeera. 2014. "Boko Haram seizes another Nigerian town."
- Amnesty International. 2015. "'Our job is to shoot, slaughter and kill'. Boko Haram's reign of terror in north-east Nigeria."
- Anadolu Agency. 2015. "Boko Haram seizes Nigeria's Askira Uba."
- Angrist, Joshua D, and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Anugwom, Edlyne Eze. 2018. *The Boko Haram insurgency in Nigeria: perspectives from within*. Springer.
- Archibong, Belinda, Vijay Modi, and Shaky Sherpa. 2015. "Geography of infrastructure functionality at schools in Nigeria: Evidence from spatial data analysis across local government areas." *Papers in Applied Geography* 1 (2): 176–183.
- Arjona, Ana. 2016. "Institutions, civilian resistance, and wartime social order: a process-driven natural experiment in the Colombian Civil War." *Latin American Politics and Society* 58 (3): 99–122.
- Arjona, Ana. 2017. *Rebelocracy: A theory of social order in civil war*. Vol. 422. Helen Kellogg Institute for International Studies.
- BBC. 2015. "Nigeria's Boko Haram: Baga destruction 'shown in images'."
- Berman, Eli. 2003. *Hamas, Taliban and the Jewish underground: an economist's view of radical religious militias*.
- Bertoni, Eleonora, Michele Di Maio, Vasco Molini, and Roberto Nistico. 2019. "Education is forbidden: The effect of the Boko Haram conflict on education in North-East Nigeria." *Journal of Development Economics* 141:102249.
- Bove, Vincenzo, Jessica Di Salvatore, Leandro Elia, and Roberto Nisticò. 2024. "Mothers at peace: International peacebuilding and post-conflict fertility." *Journal of Development Economics* 167:103226.
- Brown, Ryan, Verónica Montalva, Duncan Thomas, and Andrea Velásquez. 2019. "Impact of violent crime on risk aversion: Evidence from the Mexican drug war." *The Review of Economics and Statistics* 101 (5): 892–904.
- Cameron, A Colin, Jonah B Gelbach, and Douglas L Miller. 2008. "Bootstrap-based improvements for inference with clustered errors." *The Review of Economics and Statistics* 90 (3): 414–427.

- Center for International Security and Cooperation. 2018. "Mapping Militant Organizations: "Boko Haram"." <https://cisac.fsi.stanford.edu/mappingmilitants/profiles/boko-haram>.
- CFR. 2018. "Boko Haram's Deadly Impact."
- CFR. 2022. "Conflict with Boko Haram in Nigeria."
- Conley, Timothy G. 1999. "GMM estimation with cross sectional dependence." *Journal of econometrics* 92 (1): 1–45.
- Conley, Timothy G. 2008. "Spatial econometrics," 303–313.
- Crisis group. 2016. "Exploiting Disorder: al-Qaeda and the Islamic State."
- Crisis group. 2017. "Instruments of Pain (IV): The Food Crisis in North East Nigeria."
- Di Maio, Michele, and Roberto Nisticò. 2019. "The effect of parental job loss on child school dropout: Evidence from the Occupied Palestinian Territories." *Journal of Development Economics* 141:102375.
- France24. 2014. "Nigerian forces launch assault on Boko Haram in Baga."
- GCPEA. 2020. "Global Coalition to Protect Education from Attack, 2020 Report."
- Guerra-Cújar, María Eugenia, Mounu Prem, Paul Rodríguez-Lesmes, and Juan F Vargas. 2024. "A peace baby boom? Evidence from Colombia's peace agreement." *Journal of Human Resources*, 1222–12701R1.
- Ho, Daniel E, Kosuke Imai, Gary King, and Elizabeth A Stuart. 2007. "Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference." *Political analysis* 15 (3): 199–236.
- IOM. 2015. "Nigeria crisis. IOM regional assessment findings and recommendations."
- Iyekekpolo, Wisdom Oghosa. 2016. "Boko Haram: understanding the context." *Third World Quarterly* 37 (12): 2211–2228.
- Justino, Patricia. 2011. "Violent conflict and human capital accumulation." *IDS Working Papers* 2011 (379): 1–17.
- Kalyvas, Stathis N. 2006. *The logic of violence in civil war*. Cambridge University Press.
- Mampilly, Zachariah. 2021. "Counting school children under rebel control: four case studies."
- Maynard, Jonathan Leader. 2019. "Ideology and armed conflict." *Journal of Peace Research* 56 (5): 635–649.

- Omenma, J Tochukwu, Ifeanyichukwu M Abada, and Z Onyinyechi Omenma. 2020. "Boko Haram insurgency: A decade of dynamic evolution and struggle for a caliphate." *Security Journal*, 1–25.
- Peters, Michael A. 2014. 'Western education is sinful': Boko Haram and the abduction of Chibok schoolgirls, 2.
- Raleigh, Clionadh, Andrew Linke, Håvard Hegre, and Joakim Karlsen. 2010. "Introducing ACLED-Armed Conflict Location and Event Data." *Journal of Peace Research* 47 (5): 651–660.
- Reliefweb. 2017. "Operational Presence by Education Core Commitment."
- Roodman, David, Morten Ørregaard Nielsen, James G MacKinnon, and Matthew D Webb. 2019. "Fast and wild: Bootstrap inference in Stata using boottest." *The Stata Journal* 19 (1): 4–60.
- Rosenbaum, Paul R, PR Rosenbaum, and Briskman. 2010. *Design of observational studies*. Vol. 10. Springer.
- Rubin, Donald B. 2007. "The design versus the analysis of observational studies for causal effects: parallels with the design of randomized trials." *Statistics in medicine* 26 (1): 20–36.
- Rubin, Michael A. 2020. "Rebel territorial control and civilian collective action in civil war: Evidence from the communist insurgency in the Philippines." *Journal of Conflict Resolution* 64 (2-3): 459–489.
- Shemyakina, Olga. 2011. "The effect of armed conflict on accumulation of schooling: Results from Tajikistan." *Journal of Development Economics* 95 (2): 186–200.
- Stewart, Megan A. 2018. "Civil war as state-making: Strategic governance in civil war." *International Organization* 72 (1): 205–226.
- Sundberg, Ralph, and Erik Melander. 2013. "Introducing the UCDP georeferenced event dataset." *Journal of peace research* 50 (4): 523–532.
- Swee, Eik Leong. 2015. "On war intensity and schooling attainment: The case of Bosnia and Herzegovina." *European Journal of Political Economy* 40:158–172.
- Thurston, Alex. 2016. "'The disease is unbelief': Boko Haram's religious and political worldview."
- Unicef. 2015. "Nigeria. Humanitarian Situation Report."
- Wood, Elisabeth Jean. 2008. "The social processes of civil war: The wartime transformation of social networks." *Annu. Rev. Polit. Sci.* 11:539–561.
- Yakter, Alon, and Liran Harsgor. 2022. "Long-term Change in Conflict Attitudes: A Dynamic Perspective." *British Journal of Political Science*, 1–19.

Beyond violence: The educational consequences of insurgent rule

Online Appendix

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Appendix A Background of Boko Haram

Boko Haram is one of the largest militant groups in Africa, primarily active around the Lake Chad Basin in northeast Nigeria (CFR 2022). For a comprehensive review of the group's background and ideology, see Iyekekpola (2016). In summary, "the Boko Haram sect has advanced the message of reform and purification of the Nigerian political system against Western values and culture, which it argues are the cause of corruption and economic hardship" (p. 2213). Muhammed Yusuf, a key leader during the period considered in this study, "advocated a withdrawal from the secular system of the Nigerian state and showed discontent with Western education. He believed that the Nigerian system represented the Western education model (boko), which must be discontinued, as it was forbidden for Muslims—hence the nickname given to his movement: Boko Haram" (p. 2218) (Iyekekpola 2016).

The insurgency began in 2009 and escalated rapidly in the following years. By spring 2014, Boko Haram controlled several LGAs and declared a caliphate in August of that year.¹ According to interviews conducted by Amnesty International (2015) with former residents of the caliphate, Boko Haram enforced governance consistent with its strong anti-democratic and anti-Western education stance. Central to its rule was the prohibition of formal schooling, in line with its broader rejection of secular elements of Nigerian society (Center for International Security and Cooperation 2018; Omenma et al. 2020; Anugwom 2018; Thurston 2016; CFR 2018).

However, the vastness of the territory under Boko Haram's control made uniform rule enforcement difficult. This led to discrepancies between the group's stated policies and their implementation. In some villages, civilians were imprisoned or under constant surveillance, while in others, residents retained some freedom of movement—though fighters frequently patrolled both streets and routes between villages to prevent escape (Amnesty International 2015). For this study, this suggests that while schooling was officially banned, limited education may have continued in certain areas.

In January 2015, a regional task force launched a counterinsurgency operation that culminated in the recapture of Gwoza, the de facto headquarters of Boko Haram, in March 2015, effectively ending the territorial control of the group.² After the group was expelled, violence in the northeast declined, though Boko Haram continued to carry out attacks (CFR 2018). The region nonetheless faced long-lasting challenges, including widespread infrastructure destruction and trade restrictions aimed at cutting off insurgent supply lines, which contributed to heightened food insecurity (Crisis group 2017). These conditions were particularly severe in Borno, Adamawa, and Yobe states.

Following the government's restoration of territorial control, humanitarian agencies began delivering assistance, including food aid, healthcare, and educational support. In all previously occupied states included in the study sample, NGOs launched emergency learning initiatives and worked to

1. Boko Haram continued to expand its territory into early 2015. Figure I.6 shows the progression of the occupation of the group, based on ACLED data. Its spread was largely restricted to peripheral regions of Borno State, likely avoiding central areas due to the presence of government forces stationed in Maiduguri (the capital) and nearby areas as of September 2014 (OCHA).

2. See Figure I.5.

strengthen the educational system (Reliefweb 2017).

Appendix B Dropouts

I examine which children are most likely to permanently drop out of school, defined as not returning to school at any point during the six years following the occupation. Table B.1 compares the characteristics of children who drop out with those who remain enrolled. Dropouts are marginally more likely to come from Muslim, smaller, and rural households but are exposed to slightly less violence and lower levels of conflict risk. All these variables are included as covariates in the main analysis. Table B.2 further compares dropouts across treatment status. Children who drop out in the control group tend to come from somewhat larger households, are on average about one year older, and are slightly less likely to be of school age at the time of treatment—likely reflecting the correlation between age and school eligibility. While children in treated areas are exposed to somewhat higher levels of conflict risk, those in control areas experience marginally more violence. Overall, these differences are small in magnitude and do not suggest systematic selection that would threaten the validity of the estimated treatment effects.

Table B.1: Not returning to school: stayers vs. drop-outs

	Stayers	Dropouts	Difference	t-statistic	N
Age	13.841	14.641	-0.800***	-3.05	792
School-going age	0.393	0.327	0.066*	1.90	792
Female	0.393	0.382	0.010	0.29	792
Muslim	0.570	0.707	-0.137***	-4.01	792
Household size	11.704	10.788	0.916***	2.63	792
Agricultural household	0.202	0.210	-0.008	-0.26	792
Rural	0.816	0.911	-0.095***	-3.95	792
Conflict risk	1.871	1.383	0.488***	3.74	792
Boko Haram events	1.174	0.259	0.915***	2.69	792

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Not returning to school: control vs. treatment group

	Control	Treatment	Difference	t-statistic	N
Age	14.867	13.411	1.456***	3.04	471
School-going age	0.276	0.603	-0.326***	-5.63	471
Female	0.369	0.452	-0.083	-1.34	471
Muslim	0.749	0.479	0.269***	4.75	471
Household size	11.191	8.589	2.602***	4.39	471
Agricultural household	0.206	0.233	-0.027	-0.52	471
Rural	0.894	1.000	-0.106***	-2.93	471
Conflict risk	1.279	1.948	-0.669***	-2.99	471
Boko Haram events	0.307	0.000	0.307**	2.16	471

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C Data and Summary Statistics

Table C.1: Sample size per school-year

School-year	Control	Treatment	Total
2009/10	210	49	259
2010/11	218	53	271
2011/12	178	40	218
2012/13	178	55	233
2015/16	221	59	280
2017/18	119	17	136
2018/19	119	17	136
2020/21	67	2	69
2022/23	79	14	93
2023/24	68	10	78
Total	1457	316	1773

Notes: This table reports the number of children observed in each school-year by treatment status.

Table C.2: Descriptive statistics

	Mean	SD	Min.	Max.	N
Age	6.65	2.49	2	12	981
School-going age	0.78	0.41	0	1	981
Female	0.44	0.50	0	1	981
Muslim	0.72	0.45	0	1	981
Household size	9.08	3.70	3	26	981
Agricultural household	0.25	0.43	0	1	981
Hh. business	0.01	0.11	0	1	981
Hh. Farm	0.13	0.33	0	1	981
External job	0.01	0.09	0	1	981
Ill/Injured	0.08	0.27	0	1	495
Visited doctor	0.07	0.25	0	1	495
Marriage	0.01	0.07	0	1	981
Household/childcare	0.03	0.17	0	1	981
Parents' disinterest	0.11	0.31	0	1	981
Too old for school	0.00	0.06	0	1	981
No schools/teachers	0.03	0.16	0	1	981
Primary school	0.96	0.21	0	1	981
Secondary school	0.88	0.32	0	1	981
Boko Haram events	1.93	10.71	0	96	981
Boko Haram fatalities	11.79	57.06	0	503	981
Boko Haram events (UCDP)	1.13	8.00	0	102	981
Boko Haram events (UCDP)	6.81	56.26	0	912	981
State-involved conflict	1.05	5.70	0	48	981
State-involved fatalities	8.92	68.44	0	854	981
Other actor conflict	0.31	1.39	0	11	981
Other actor fatalities	0.43	2.43	0	20	981
Conflict risk	4.38	9.20	0	57	981
Attendance	0.58	0.49	0	1	980
Years of educ.	1.74	2.30	0	10	980

Table C.3: Control and treatment group, pre-treatment

	Control	Treatment	Difference	t-statistic	N
Age	6.765	6.213	0.552***	2.80	981
School-going age	0.796	0.741	0.055*	1.67	981
Female	0.441	0.452	-0.010	-0.26	981
Muslim	0.723	0.706	0.018	0.49	981
Household size	9.454	7.574	1.880***	6.50	981
Agricultural household	0.233	0.315	-0.081**	-2.36	981
Rural	0.756	1.000	-0.244***	-7.96	981
Conflict risk	4.994	1.958	3.036***	4.17	981
Boko Haram events	2.401	0.071	2.329***	2.74	981
Boko Haram events (UCDP)	1.401	0.076	1.324**	2.08	981

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Control and treatment group, pre-treatment (LGA)

	Control	Treatment	Difference	t-statistic	N
Primary school	0.940	1.000	-0.060***	-2.98	116
Secondary school	0.848	0.941	-0.094**	-2.52	116
Bus stop	0.548	0.741	-0.193*	-1.76	100
Post office	0.191	0.426	-0.235***	-2.70	108
Market	0.574	0.885	-0.311***	-3.40	110
Population density	617.879	76.994	540.885**	2.18	116
Hospital	0.373	0.556	-0.182*	-1.86	108
Health center	0.653	1.000	-0.347***	-4.77	110
Boko Haram events (ACLED)	3.028	0.075	2.953	1.26	116
Boko Haram fatalities (ACLED)	18.085	0.303	17.782	1.40	116
Boko Haram events (UCDP)	1.200	0.057	1.143	0.88	116
Boko Haram fatalities (UCDP)	7.585	0.569	7.016	0.86	116
Conflict event: battles	5.211	1.148	4.063**	2.00	116
Conflict event: explosions	3.010	0.185	2.825**	2.26	116
Conflict event: attacks on civilians	5.745	1.422	4.323**	2.01	116
Conflict risk	5.164	2.124	3.040	1.61	116
Support for Boko Haram	0.513	0.609	-0.095	-0.86	85
Distance to Sambisa forest (km)	163.727	150.897	12.831	0.90	106
Muslim	0.701	0.690	0.011	0.12	116
Agricultural household	0.236	0.276	-0.039	-0.43	116
Rural area	0.793	1.000	-0.207***	-2.73	116

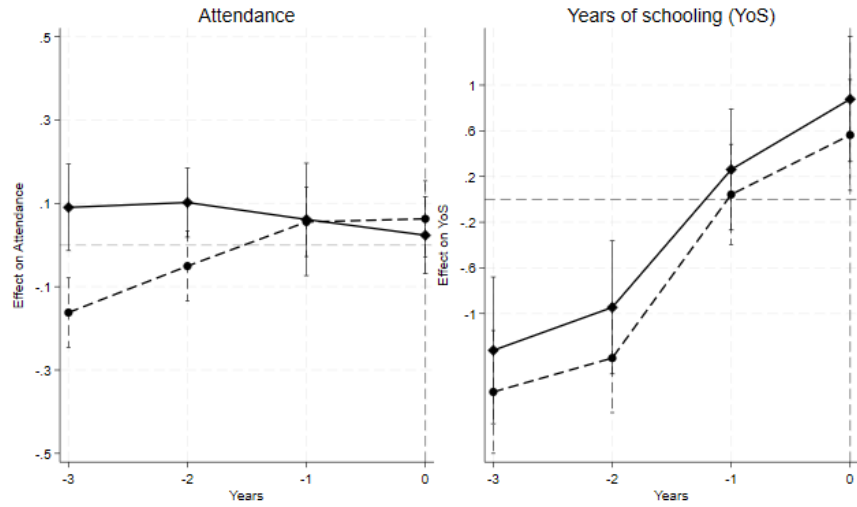
Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6: Attrition among school-age children

	(1)
treat	-0.002 (0.003)
Conflict risk	0.000 (0.000)
Boko Haram events	-0.000 (0.000)
Agricultural household	-0.018 (0.018)
Household size	0.002 (0.002)
Rural	0.015 (0.015)
Female	0.011 (0.011)
School-going age	-0.014 (0.014)
Muslim	0.020 (0.020)
Age	0.001 (0.001)
Observations	1773

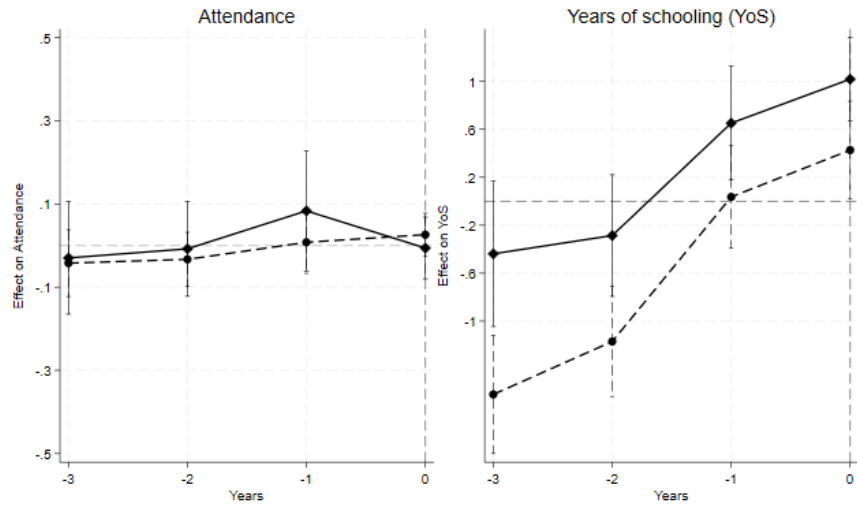
Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors clustered at the village level. Attrition is defined as dropping out of the panel before aging out of school (ages 5–15). The coefficient on treatment is small and statistically insignificant, indicating no systematic attrition by treatment status.

Figure C.1: Pre-event study: female



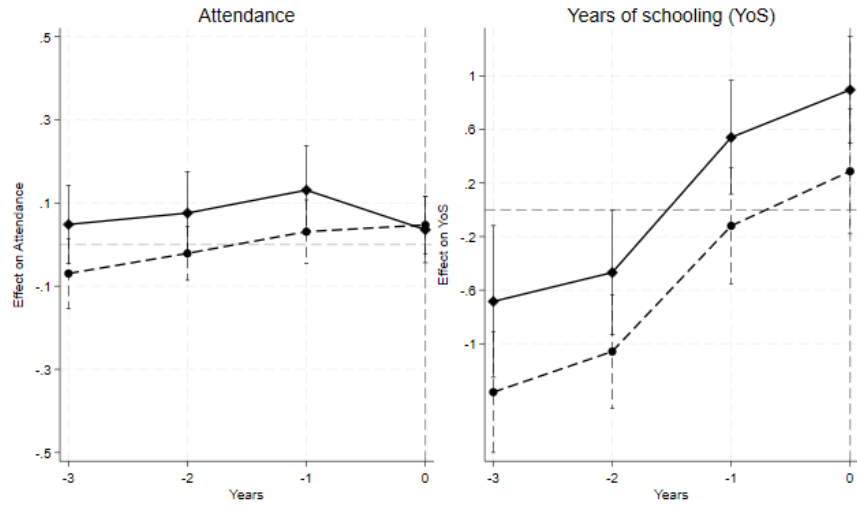
Note: The trends for the treatment and control group, for female children, for school attendance (left) and years of education (right). Treatment (solid line) and control group (dashed line). 95% confidence intervals shown.

Figure C.2: Pre-event study: male



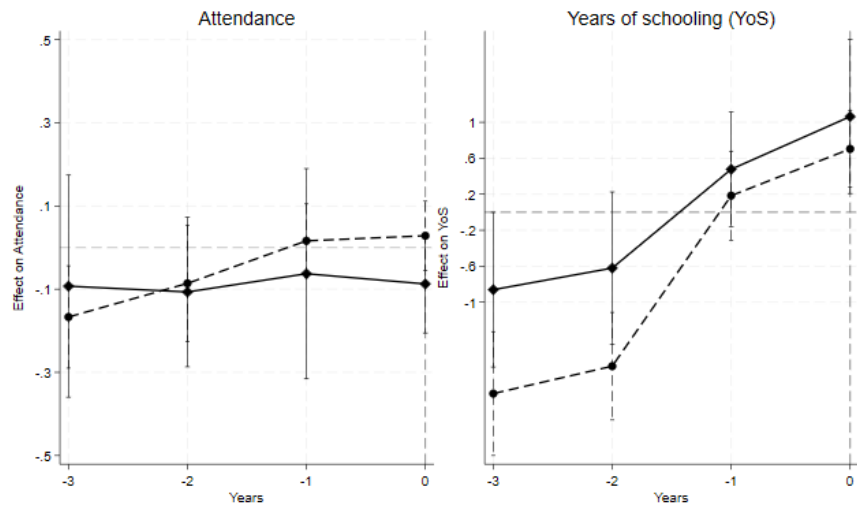
Note: The trends for the treatment and control group, for male children, for school attendance (left) and years of education (right). Treatment (solid line) and control group (dashed line). 95% confidence intervals shown.

Figure C.3: Pre-event study: Muslim household



Note: The trends for the treatment and control group, for children of Muslim households, for school attendance (left) and years of education (right). Treatment (solid line) and control group (dashed line). 95% confidence intervals shown.

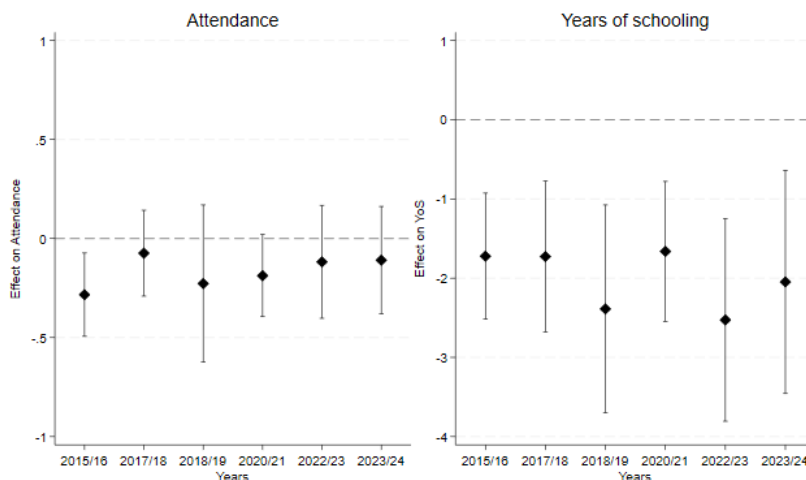
Figure C.4: Pre-event study: non-Muslim household



Note: The trends for the treatment and control group, for children of non-Muslim households, for school attendance (left) and years of education (right). Treatment (solid line) and control group (dashed line). 95% confidence intervals shown.

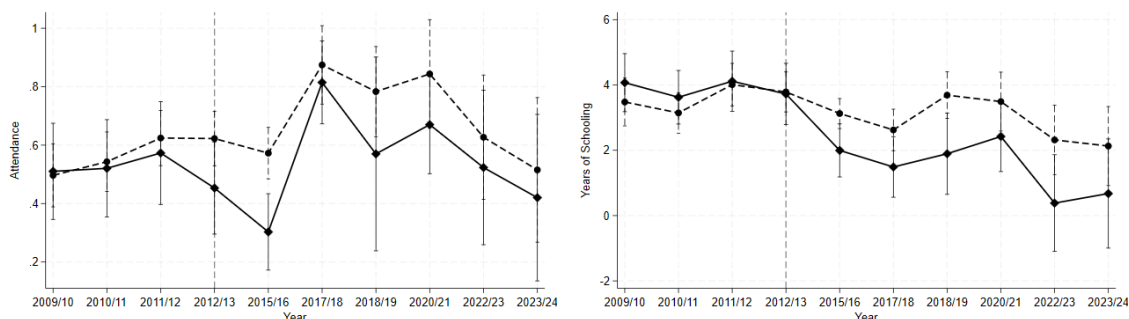
Appendix D Results

Figure D.1: Estimated coefficients per school year



Notes: The estimated treatment effects for school attendance (left panel) and YoS (right panel), by school year. 95% confidence intervals shown.

Figure D.2: Attendance, treatment and control group

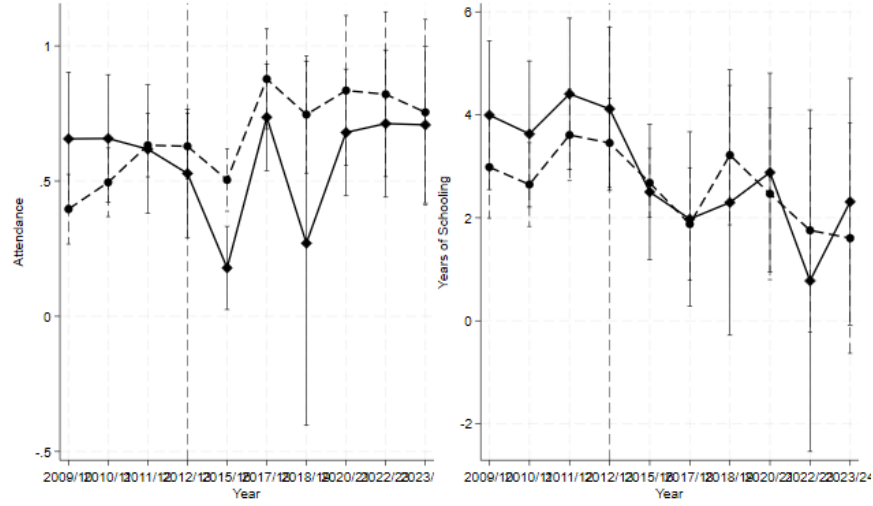


Note: The trends for the treatment and control group for school attendance (left) and years of education (right) over full time period. Treatment (solid line) and control group (dashed line). 95% confidence intervals shown.

Subsection D.1 Violence, occupation, and conflict risk

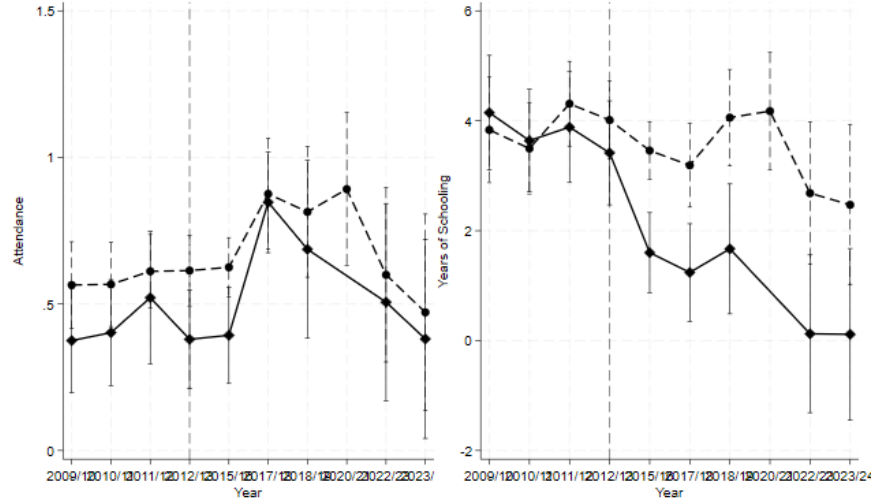
To further disentangle the sources of educational disruption, Table D.1 incrementally adds violence exposure, occupation, and conflict risk. Violence alone has no significant effect on years of schooling or attendance. By contrast, occupation has a large and significant negative effect, even when controlling for both violence and conflict risk. The conflict risk coefficient is positive and significant for years of schooling, indicating that children farther from Boko Haram's stronghold have better outcomes—consistent with a threat gradient.

Figure D.3: Event study: attendance and YoS, female



Note: The trends for the treatment and control group for school attendance (left) and years of education (right) over full time period. Treatment (solid line) and control group (dashed line). 95% confidence intervals shown.

Figure D.4: Event study: attendance and YoS, male



Note: The trends for the treatment and control group for school attendance (left) and years of education (right) over full time period. Treatment (solid line) and control group (dashed line). 95% confidence intervals shown.

Table D.1: Violence, occupation, and conflict risk

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Boko Haram events	0.0129*** (0.0038)	0.0134*** (0.0038)	0.0134*** (0.0036)	-0.0077 (0.0077)	-0.0068 (0.0077)	-0.0060 (0.0079)	-0.0010 (0.0018)	-0.0009 (0.0019)	-0.0010 (0.0018)
Occupied		-0.7636*** (0.1902)	-0.7416*** (0.1907)		-1.3861*** (0.3433)	-1.4805*** (0.3447)		-0.1632* (0.0779)	-0.1526* (0.0763)
Conflict risk			-0.0095 (0.0197)			0.0253* (0.0124)			-0.0028 (0.0017)
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	1,447	1,447	1,758	1,758	1,758	1,647	1,647	1,647
R-squared	0.009	0.019	0.019	0.030	0.053	0.059	0.029	0.035	0.036

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time lag of three school-years. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. All estimations include individual and school-year fixed effects. Controls include household size, agricultural household status, child's school-age status, and whether a state of emergency was declared prior to the occupation.

Subsection D.2 Nesting exercise: occupation effects beyond violence

Table D.2 reports a nesting exercise that incrementally adds the occupation treatment to a specification based on a Bertoni-style violence measure (Boko Haram events within 10km of the household). The occupation coefficient remains large and significant after controlling for violence, confirming that the effects documented here capture something distinct from and additional to the effects of violence exposure.

Table D.2: Disentangling Violence from Occupation (Nesting Bertoni et al. (2019))

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Occupied		-0.7636*** (0.1902)	-0.7416*** (0.1907)		-1.3861*** (0.3433)	-1.4805*** (0.3447)		-0.1632** (0.0779)	-0.1526** (0.0763)
Boko Haram events	0.0129*** (0.0038)	0.0134*** (0.0038)	0.0134*** (0.0036)	-0.0077 (0.0077)	-0.0068 (0.0077)	-0.0060 (0.0079)	-0.0010 (0.0018)	-0.0009 (0.0019)	-0.0010 (0.0018)
Conflict risk			-0.0095 (0.0197)			0.0253** (0.0124)			-0.0028 (0.0017)
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	1,447	1,447	1,758	1,758	1,758	1,647	1,647	1,647
R-squared	0.009	0.019	0.019	0.030	0.053	0.059	0.029	0.035	0.036

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. Progressive nesting: columns (1)/(4)/(7) include violence exposure only; (2)/(5)/(8) add occupation treatment; (3)/(6)/(9) add conflict risk. All estimations include individual and school-year fixed effects. Controls: household size; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

Appendix E Mechanisms

Table E.1: Mechanisms: child labor

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Hh. business	Hh. business	Hh. business	Hh. Farm	Hh. Farm	Hh. Farm	External job	External job	External job
Occupied	-0.0207 (0.0267)	-0.0317 (0.0290)	-0.0504* (0.0212)	-0.0241 (0.0926)	0.0012 (0.1149)	-0.1438 (0.0936)	0.0000 (0.0071)	-0.0014 (0.0075)	-0.0086 (0.0106)
Occupation*Female		0.0259 (0.0226)			-0.0594 (0.0893)			0.0033 (0.0025)	
Occupation*Muslim			0.0529 (0.0397)			0.2124 (0.1475)			0.0152 (0.0102)
Boko Haram events	-0.0001 (0.0004)	-0.0001 (0.0004)	-0.0001 (0.0004)	0.0006 (0.0015)	0.0006 (0.0015)	0.0007 (0.0015)	-0.0005 (0.0004)	-0.0005 (0.0004)	-0.0005 (0.0004)
Conflict risk	0.0024** (0.0008)	0.0024** (0.0008)	0.0024** (0.0008)	0.0015 (0.0020)	0.0015 (0.0020)	0.0014 (0.0019)	-0.0010 (0.0006)	-0.0010 (0.0006)	-0.0010 (0.0006)
Mean(Y)	0.017	0.017	0.017	0.144	0.144	0.144	0.007	0.007	0.007
Observations	1,554	1,554	1,554	1,554	1,554	1,554	1,554	1,554	1,554
R-squared	0.011	0.011	0.012	0.003	0.004	0.008	0.006	0.006	0.006

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior and a measure for conflict risk; household size; whether it is an agricultural household; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

Table E.2: Mechanisms: health and marriage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Ill/Injured	Ill/Injured	Ill/Injured	Visited doctor	Visited doctor	Visited doctor	Marriage	Marriage	Marriage
Occupied	0.0086 (0.0409)	0.0046 (0.0505)	-0.0669 (0.0822)	0.0104 (0.0368)	0.0213 (0.0425)	-0.0570 (0.0732)	-0.0074 (0.0088)	0.0045 (0.0036)	0.0051 (0.0039)
Occupation*Female		0.0086 (0.0388)			-0.0231 (0.0284)			-0.0265 (0.0174)	
Occupation*Muslim			0.0996 (0.0795)			0.0901 (0.0702)			-0.0168 (0.0087)
Boko Haram events	0.0007 (0.0008)	0.0007 (0.0008)	0.0007 (0.0008)	0.0006 (0.0007)	0.0006 (0.0007)	0.0006 (0.0007)	-0.0000 (0.0001)	0.0000 (0.0001)	-0.0000 (0.0001)
Conflict risk	0.0010 (0.0020)	0.0010 (0.0020)	0.0011 (0.0020)	0.0027 (0.0025)	0.0027 (0.0025)	0.0028 (0.0025)	-0.0003 (0.0002)	-0.0003 (0.0002)	-0.0003 (0.0002)
Mean(Y)	0.087	0.087	0.087	0.088	0.088	0.088	0.006	0.006	0.006
Observations	748	748	748	759	759	759	1,263	1,263	1,263
R-squared	0.006	0.006	0.008	0.010	0.010	0.012	0.002	0.004	0.003

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior and a measure for conflict risk; household size; whether it is an agricultural household; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

Table E.3: Mechanisms: household or childcare, parental disinterest, age

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Household/childcare	Household/childcare	Household/childcare	Parents' disinterest	Parents' disinterest	Parents' disinterest	Too old for school	Too old for school	Too old for school
Occupied	-0.0188 (0.0185)	-0.0159 (0.0199)	-0.0027 (0.0211)	0.1192* (0.0494)	0.0495 (0.0446)	0.0993** (0.0375)	-0.0069 (0.0147)	-0.0035 (0.0188)	0.0079 (0.0132)
Occupation*Female		-0.0068 (0.0149)			0.1623* (0.0711)			-0.0079 (0.0100)	
Occupation*Muslim			-0.0299 (0.0294)			0.0364 (0.0858)			-0.0273 (0.0239)
Boko Haram events	0.0025 (0.0025)	0.0025 (0.0025)	0.0025 (0.0025)	-0.0004 (0.0009)	-0.0004 (0.0009)	-0.0004 (0.0009)	0.0001 (0.0001)	0.0001 (0.0001)	0.0000 (0.0001)
Conflict risk	0.0010 (0.0007)	0.0010 (0.0007)	0.0010 (0.0007)	0.0030* (0.0013)	0.0029* (0.0013)	0.0029* (0.0013)	0.0000 (0.0002)	0.0000 (0.0002)	0.0000 (0.0002)
Mean(Y)	0.018	0.018	0.018	0.100	0.100	0.100	0.008	0.008	0.008
Observations	1,773	1,773	1,773	1,758	1,758	1,758	1,773	1,773	1,773
R-squared	0.010	0.010	0.010	0.014	0.017	0.014	0.005	0.005	0.006

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior and a measure for conflict risk; household size; whether it is an agricultural household; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

Table E.4: Mechanisms: education premiums

	(1)	(2)	(3)	(4)	(5)	(6)
	Employed	Employed	Employed	Educ. premium	Educ. premium	Educ. premium
Occupation	0.0170 (0.0651)	0.0041 (0.0788)	-0.0261 (0.0707)	0.1445** (0.0533)	0.0590 (0.0664)	0.1440* (0.0591)
Occupation*Female		0.0555 (0.0881)			0.1764** (0.0592)	
Occupation*Muslim			0.0876 (0.0893)			0.0009 (0.0636)
Boko Haram violence	-0.0008 (0.0019)	-0.0009 (0.0018)	-0.0008 (0.0019)	-0.0029* (0.0013)	-0.0027* (0.0013)	-0.0029* (0.0013)
Conflict risk	-0.0025 (0.0019)	-0.0028 (0.0019)	-0.0025 (0.0019)	-0.0030 (0.0022)	-0.0030 (0.0022)	-0.0030 (0.0022)
Mean(Y)	0.553	0.553	0.553	0.373	0.373	0.373
Observations	2,130	1,912	2,130	2,130	1,912	2,130
R-squared	0.030	0.031	0.031	0.018	0.018	0.018

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. Sample only includes respondents born before 1998 from households of which children from the original sample. Sample in column 4-6 is restricted to adults who indicate to be employed. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior; household size; whether it is an agricultural household; whether the respondent is female; whether respondent is Muslim; a dummy that captures whether the state of emergency was declared prior to the occupation.

Table E.5: Mechanisms: schools and teachers

	(1)	(2)	(3)
	No schools/teachers	Primary school	Secondary school
Occupied	-0.0579 (0.0482)	-0.0360 (0.0377)	-0.1151 (0.0739)
Boko Haram events	0.0001 (0.0004)	0.0009 (0.0009)	-0.0013 (0.0026)
Conflict risk	0.0001 (0.0006)	0.0021 (0.0016)	0.0114** (0.0039)
Mean(Y)	0.02	0.97	0.91
Observations	1,758	1,261	1,261
R-squared	0.007	0.027	0.063

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior; household size; whether it is an agricultural household; whether the child is female; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

Appendix F Additional Inference and Robustness

Subsection F.1 Wild cluster bootstrap

Given the small number of treated clusters (four occupied LGAs), standard inference based on cluster-robust standard errors may be unreliable. Table F.1 reports results from a wild cluster bootstrap procedure following Cameron et al. (2008) and Roodman et al. (2019), clustering at the LGA level with 9,999 bootstrap replications. The main results remain significant under this more conservative inference procedure.

Table F.1: Wild Cluster Bootstrap Inference

	(1)	(2)	(3)
	Increase YoS	Years of educ.	Attendance
Occupied	-0.7416 (0.3399)	-1.4805 (0.5232)	-0.1526 (0.1213)
Bootstrap p -value	[0.0309]	[0.0360]	[0.1611]
Bootstrap 95% CI	[-1.638, -0.083]	[-2.689, -0.163]	[-0.745, 0.054]
Observations	1,447	1,758	1,647

Notes: Wild cluster bootstrap with 9,999 replications (Roodman et al., 2019). Clustering at the LGA level. Controls: household size, school age, state of emergency, BH events within 10km, conflict risk.

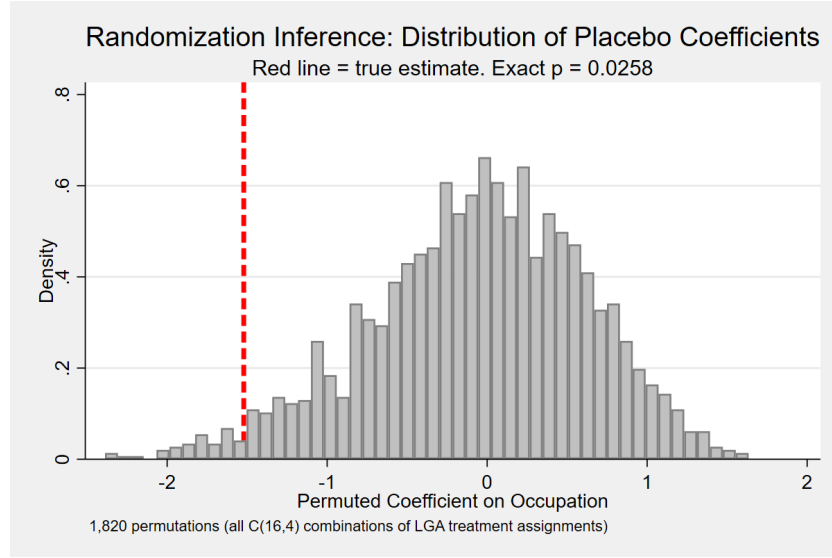
Subsection F.2 Randomization inference

Figure F.1 presents the distribution of placebo treatment effects from an exact randomization inference exercise. I enumerate all $\binom{16}{4} = 1,820$ possible assignments of four treated LGAs from the sixteen LGAs in the sample, re-estimate the baseline specification for each permutation, and record the resulting placebo coefficient. The observed treatment effect is compared to this permutation distribution. The exact p -value is calculated as the share of permuted coefficients at least as extreme as the actual estimate in absolute value.

Subsection F.3 Leave-one-LGA-out

Table F.2 reports results from sequentially dropping each of the four occupied LGAs from the sample. The main results are robust to the exclusion of any single treated area, indicating that the findings are not driven by any particular occupied LGA.

Figure F.1: Randomization inference: permutation distribution of placebo treatment effects



Note: The figure shows the distribution of 1,820 placebo treatment effects from an exact enumeration of all $\binom{16}{4}$ possible assignments of four treated LGAs. The vertical red line indicates the observed treatment effect.

Table F.2: Leave-One-Treated-LGA-Out Robustness

	Full sample	Drop LGA 210	Drop LGA 815	Drop LGA 824	Drop LGA 3506
<i>Panel A: Increase in YoS</i>					
Occupied	-0.7416	-1.1653	-0.7141	-0.6960	-0.5263
	(0.1907)	(0.2095)	(0.2043)	(0.2031)	(0.1899)
Observations	1,447	1,331	1,409	1,415	1,381
<i>Panel B: Years of educ.</i>					
Occupied	-1.4805	-1.6839	-1.5611	-1.5555	-1.0990
	(0.3447)	(0.4363)	(0.3591)	(0.3632)	(0.3031)
Observations	1,758	1,621	1,709	1,717	1,671
<i>Panel C: Attendance</i>					
Occupied	-0.1526	-0.2457	-0.0673	-0.1713	-0.1728
	(0.0763)	(0.1163)	(0.0447)	(0.0832)	(0.1063)
Observations	1,647	1,523	1,598	1,607	1,558

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each column drops one treated LGA from the sample. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. Individual and school-year fixed effects. Controls: household size, school age, state of emergency, BH events within 10km, conflict risk.

Subsection F.4 LGA-specific time trends

Table F.3 reports results from a specification that includes LGA-specific linear time trends. This addresses the concern that occupied and non-occupied LGAs may have been on different schooling trajectories even absent occupation. Adding LGA trends attenuates the occupation coefficient on cumulative years of schooling and attendance, but the effect on annual grade progression remains large and, if anything, increases in magnitude. All three outcomes retain the expected negative sign.

Table F.3: Robustness to LGA-Specific Linear Time Trends

	(1)	(2)	(3)	(4)	(5)	(6)
	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Attendance	Attendance
Occupied	-0.7416*** (0.1907)	-0.8843*** (0.2891)	-1.4805*** (0.3447)	-0.7331** (0.2931)	-0.1526** (0.0763)	-0.1000 (0.1028)
Boko Haram events	0.0134*** (0.0036)	0.0164*** (0.0052)	-0.0060 (0.0079)	-0.0023 (0.0069)	-0.0010 (0.0018)	-0.0002 (0.0013)
Conflict risk	-0.0095 (0.0197)	-0.0035 (0.0246)	0.0253** (0.0124)	-0.0162 (0.0164)	-0.0028 (0.0017)	0.0064* (0.0038)
Mean(Y)	0.762	0.770	3.369	3.369	0.614	0.614
Observations	1,447	1,433	1,758	1,758	1,647	1,645
R-squared	0.019	0.369	0.059	0.869	0.036	0.595

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Odd columns: baseline specification with Conley (1999, 2008) spatially adjusted standard errors (20km, 3-year lag). Even columns: add LGA-specific linear time trends with village-clustered standard errors. All estimations include individual and school-year fixed effects. Controls: household size, school age, state of emergency, BH events within 10km, conflict risk.

Subsection F.5 Baseline results without violence controls

Table 1 (shown in the main text) reports the preferred specification, which excludes violence exposure and conflict risk controls. This addresses the concern that these variables may be endogenous to the treatment—occupation is one manifestation of the same conflict process that generates violence, military operations, and displacement. Conditioning on variables that may themselves be affected by occupation risks introducing bias (Angrist and Pischke 2009). The stability of the occupation coefficient across specifications with and without violence controls provides reassurance that the results are not sensitive to the inclusion of potentially endogenous controls.

Appendix G Robustness

Subsection G.1 Measures of violence

To assess the robustness of the main findings, I examine the sensitivity of the results to alternative definitions of conflict exposure. Specifically, I vary the distance (5, 10 and 20km) used to measure exposure and distinguish between attacks perpetrated by insurgents and those by state forces, as well as those targeting schools. For the latter, I consider all Boko Haram related events that targeted the school closest to the household.³ The estimated effects remain stable across all specifications. The results are shown in table G.1 and G.2.

Table G.1: Robustness: varying distances

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Occupied	-0.8462*** (0.2054)	-0.8482*** (0.2057)	-0.8533*** (0.2068)	-1.5735*** (0.3063)	-1.5715*** (0.3068)	-1.5715*** (0.3077)	-0.1697* (0.0771)	-0.1698* (0.0773)	-0.1700* (0.0773)
Boko Haram events, 5km	0.0239 (0.0566)			0.0566 (0.0496)			0.0073 (0.0085)		
Boko Haram events		0.0129** (0.0041)			-0.0091 (0.0080)			-0.0002 (0.0019)	
Boko Haram events, 20km			0.0088 (0.0057)			-0.0029 (0.0077)			0.0001 (0.0017)
Conflict risk									
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	1,447	1,447	1,758	1,758	1,758	1,647	1,647	1,647
R-squared	0.013	0.015	0.014	0.033	0.033	0.033	0.007	0.007	0.007

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior; household size; whether it is an agricultural household; whether the child is female; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

3. The location of schools in the area come from Archibong et al. (2015).

Table G.2: Robustness: perpetrator and target

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Occupied	-0.7379*** (0.1909)	-0.7664*** (0.2080)	-0.7343*** (0.1906)	-1.4809*** (0.3447)	-1.2132** (0.4693)	-1.4854*** (0.3455)	-0.1529* (0.0764)	-0.2074* (0.0943)	-0.1512* (0.0761)
State-involved conflict	0.0175 (0.0132)			-0.0152 (0.0180)			-0.0005 (0.0041)		
School-targeted conflict		-0.0007 (0.0041)			-0.0029 (0.0057)			0.0003 (0.0005)	
Other actor conflict			-0.0306 (0.0556)			0.0192 (0.0553)			-0.0115 (0.0101)
Conflict risk	-0.0095 (0.0197)	0.0073 (0.0315)	-0.0093 (0.0198)	0.0252* (0.0124)	0.0225 (0.0351)	0.0254* (0.0123)	-0.0027 (0.0017)	0.0020 (0.0035)	-0.0027 (0.0017)
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	892	1,447	1,758	1,165	1,758	1,647	1,156	1,647
R-squared	0.019	0.079	0.018	0.059	0.095	0.059	0.036	0.036	0.037

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior; household size; whether it is an agricultural household; whether the child is female; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

I further test the sensitivity of the results by replacing the binary treatment variable with a measure that captures the duration of occupation, distinguishing between approximately half-year, three-quarter-year, and full-year occupation periods. These approximations are based on reports from IOM (2015) and ACLED. The results, shown in Table G.3, suggest that longer durations of occupation are associated with larger declines in school attendance and greater losses in YoS, consistent with expectations.

Table G.3: Robustness: duration of occupation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Duration of occupation	-0.3487*** (0.0705)	-0.3094*** (0.0867)	-0.2121*** (0.0562)	-0.3673* (0.1568)	-0.4161* (0.1895)	-0.1978 (0.1822)	-0.1025* (0.0451)	0.0125 (0.0261)	-0.0263 (0.0235)
Duration*female		-0.0791 (0.0875)			0.0969 (0.1213)			-0.2324*** (0.0255)	
Duration*Muslim			-0.2876* (0.1465)			-0.3567 (0.2797)			-0.1491** (0.0530)
Boko Haram events	0.0133*** (0.0036)	0.0133*** (0.0036)	0.0133*** (0.0037)	-0.0068 (0.0078)	-0.0068 (0.0078)	-0.0068 (0.0078)	-0.0010 (0.0018)	-0.0009 (0.0018)	-0.0010 (0.0018)
Conflict risk	-0.0093 (0.0198)	-0.0090 (0.0199)	-0.0072 (0.0200)	0.0214 (0.0123)	0.0213 (0.0123)	0.0222 (0.0123)	-0.0025 (0.0017)	-0.0022 (0.0017)	-0.0022 (0.0016)
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	1,447	1,447	1,758	1,758	1,758	1,647	1,647	1,647
R-squared	0.019	0.019	0.021	0.040	0.040	0.042	0.041	0.056	0.047

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior; household size; whether it is an agricultural household; whether the child is female; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

To assess the robustness of the results to alternative measures of violence, I re-estimate the baseline models using data from the UCDP Georeferenced Event Dataset (UCDP GED). Like

ACLED, UCDP provides detailed information on violent events, including actors, fatalities, and geographic coordinates. However, a key distinction is that UCDP only includes events with at least one reported fatality, whereas ACLED also includes non-fatal incidents. I construct household-level measures of exposure to Boko Haram violence within a 10km radius using UCDP GED, following the same procedure used for the ACLED-based measures. Results are presented in Table G.4 and remain robust to this alternative specification of violence exposure.

Table G.4: Robustness: UCDP violence data

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Occupied	-0.7378*** (0.1910)	-0.6167** (0.2074)	-0.2126 (0.1800)	-1.4796*** (0.3434)	-1.6885*** (0.3435)	-1.2097** (0.3782)	-0.1530* (0.0765)	-0.0072 (0.0581)	-0.0298 (0.0588)
Occupation*Female		-0.2886 (0.2049)			0.4937 (0.2647)			-0.3492** (0.1243)	
Occupation*Muslim			-0.9364*** (0.2562)			-0.5021 (0.4871)			-0.2147 (0.1246)
Boko Haram events (UCDP)	-0.0015 (0.0089)	-0.0015 (0.0089)	-0.0017 (0.0088)	-0.0192 (0.0106)	-0.0192 (0.0106)	-0.0193 (0.0106)	0.0005 (0.0023)	0.0005 (0.0023)	0.0004 (0.0023)
Conflict risk	-0.0093 (0.0198)	-0.0089 (0.0198)	-0.0072 (0.0198)	0.0257* (0.0123)	0.0254* (0.0124)	0.0260* (0.0123)	-0.0027 (0.0017)	-0.0025 (0.0017)	-0.0026 (0.0017)
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	1,447	1,447	1,758	1,758	1,758	1,647	1,647	1,647
R-squared	0.018	0.019	0.023	0.060	0.061	0.061	0.036	0.045	0.039

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior, based on UCDP GED; household size; whether it is an agricultural household; whether the child is female; whether the child is of school-going age; a dummy that captures whether the state of emergency was declared prior to the occupation.

Subsection G.2 Sample selection

I expand the sample to include the full North-East, significantly increasing the sample size. The results are shown in table G.5.

Table G.5: Robustness: North East sample

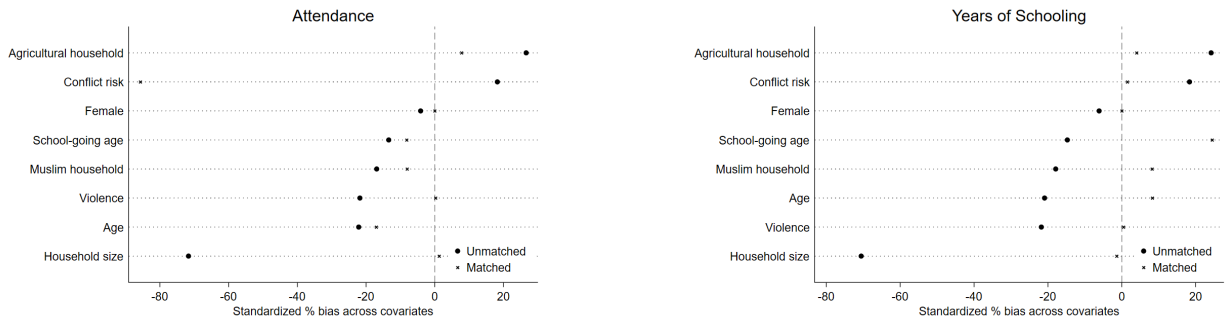
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Occupied	-0.7416*** (0.1907)	-0.6200** (0.2070)	-0.2194 (0.1785)	-1.4805*** (0.3447)	-1.6894*** (0.3447)	-1.2122** (0.3792)	-0.1526* (0.0763)	-0.0068 (0.0581)	-0.0291 (0.0587)
Occupation*Female		-0.2898 (0.2046)			0.4936 (0.2644)			-0.3492** (0.1243)	
Occupation*Muslim			-0.9310*** (0.2550)			-0.4991 (0.4878)			-0.2152 (0.1245)
Boko Haram events	0.0134*** (0.0036)	0.0134*** (0.0036)	0.0132*** (0.0037)	-0.0060 (0.0079)	-0.0060 (0.0079)	-0.0061 (0.0079)	-0.0010 (0.0018)	-0.0010 (0.0018)	-0.0011 (0.0018)
Conflict risk	-0.0095 (0.0197)	-0.0091 (0.0197)	-0.0074 (0.0197)	0.0253* (0.0124)	0.0250* (0.0124)	0.0256* (0.0124)	-0.0028 (0.0017)	-0.0025 (0.0017)	-0.0026 (0.0017)
Mean(Y)	0.762	0.762	0.762	3.369	3.369	3.369	0.614	0.614	0.614
Observations	1,447	1,447	1,447	1,758	1,758	1,758	1,647	1,647	1,647
R-squared	0.019	0.020	0.024	0.059	0.060	0.060	0.036	0.045	0.039

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. Sample includes all children living in the North East, but adhering to the other inclusion restrictions outlined in 3. All estimations include individual and school-year fixed effects. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior; household size; whether it is an agricultural household; whether the respondent is female; whether respondent is Muslim; a dummy that captures whether the state of emergency was declared prior to the occupation.

Subsection G.3 Propensity score matching

To address potential concerns regarding sample selection, I apply propensity score matching (PSM) and re-estimate the difference-in-differences model using the matched sample. This approach reduces reliance on the outcome model and helps mitigate selection bias, particularly when used in conjunction with DiD estimation (Ho et al. 2007; Rosenbaum et al. 2010; Rubin 2007). I perform nearest-neighbor matching (two neighbors, without replacement) based on all covariates included in the main specification: household size, household religion (Muslim), gender, whether the household engages in agriculture, age, and pre-treatment exposure to violence.

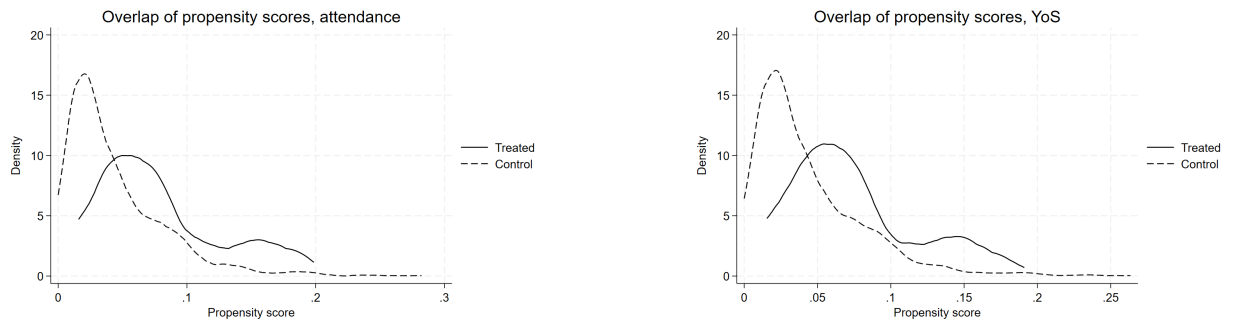
Figure G.1: Propensity score matching



Notes: The graphs above show the reduction in the bias across covariates due to matching propensity scores of observations.

The resulting distributions of propensity scores display substantial common support: virtually all observations fall within the overlapping region, and the matching algorithm retains nearly the entire sample even under restrictive caliper and no-replacement specifications (Figure G.2). This pattern reflects the nature of treatment assignment in this setting. Exposure to occupation is primarily determined by geography and timing rather than by observable household traits.

Figure G.2: Overlap of propensity scores



Notes: The graphs above show the reduction in the bias across covariates due to matching propensity scores of observations.

Given this strong common support, I restrict the robustness analysis to the region of overlap.

The finding of near-universal overlap also indicates that matching does not materially alter sample composition, reinforcing the robustness of the main results to alternative weighting and matching approaches. The results from the estimation using the PSM sample are shown in table G.6.

Table G.6: Robustness: propensity score matching

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Increase YoS	Increase YoS	Increase YoS	Years of educ.	Years of educ.	Years of educ.	Attendance	Attendance	Attendance
Occupied	-0.7434*** (0.1885)	-0.6166** (0.2094)	-0.1945 (0.1705)	-1.3058*** (0.3050)	-1.5246*** (0.3152)	-1.1160** (0.3705)	-0.1562* (0.0785)	-0.0028 (0.0565)	0.0017 (0.0606)
Occupation*Female		-0.3000 (0.2084)			0.5161 (0.2660)			-0.3659** (0.1299)	
Occupation*Muslim			-0.9685*** (0.1951)			-0.3504 (0.5019)			-0.2723* (0.1283)
Boko Haram events	0.0043 (0.0054)	0.0043 (0.0054)	0.0043 (0.0054)	-0.0005 (0.0036)	-0.0005 (0.0036)	-0.0005 (0.0036)	-0.0021 (0.0012)	-0.0021 (0.0012)	-0.0021 (0.0012)
Conflict risk	-0.0085 (0.0196)	-0.0082 (0.0196)	-0.0073 (0.0196)	0.0192 (0.0112)	0.0190 (0.0112)	0.0192 (0.0112)	-0.0024 (0.0015)	-0.0023 (0.0015)	-0.0024 (0.0015)
Mean(Y)	0.736	0.736	0.736	2.856	2.856	2.856	0.575	0.575	0.575
Observations	5,566	5,566	5,566	6,849	6,849	6,849	6,684	6,684	6,684
R-squared	0.003	0.003	0.004	0.032	0.032	0.032	0.022	0.025	0.023

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Conley (1999, 2008) spatially adjusted standard errors in parentheses: distance cutoff of 20 kilometers and a time-lag of three school-years. Sample selected through propensity score matching. All estimations include individual and school-year fixed effects. "Occupied" captures the binary treatment variable interacted with a post-2014 indicator, as in equation 1. Controls included are the number of Boko Haram related violent events within a 10km radius of the village in the school-year prior; household size; whether it is an agricultural household; whether the respondent is female; whether respondent is Muslim; a dummy that captures whether the state of emergency was declared prior to the occupation.

Appendix H Additional Tables

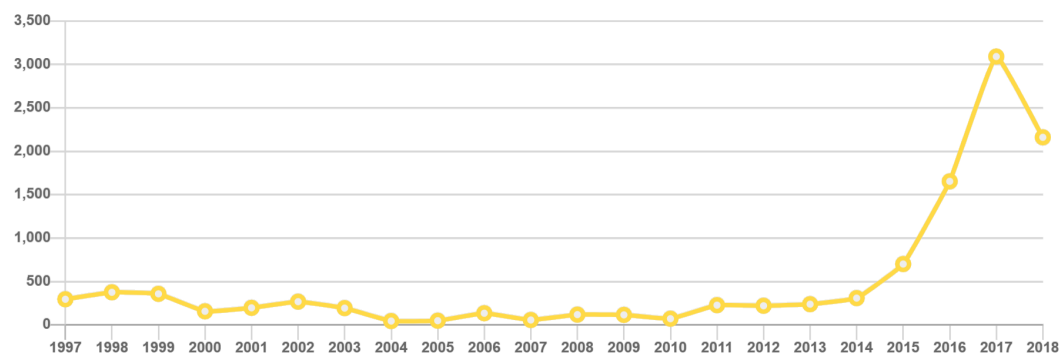
This Appendix presents additional tables regarding our analysis.

Appendix I Additional Figures

This Appendix presents additional figures regarding our analysis.

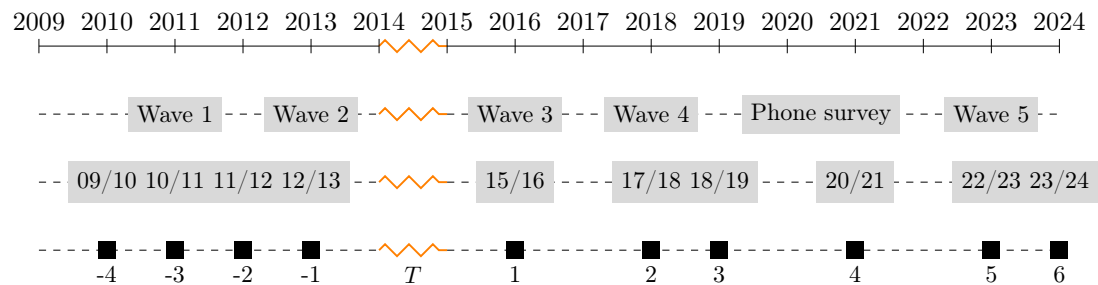
Subsection I.1 Data and trends

Figure I.1: Number of conflict events where territory is captured



Notes: The graph shows the total number of conflict events, per year, where territory was captured by non-state actors. Source: ACLED.

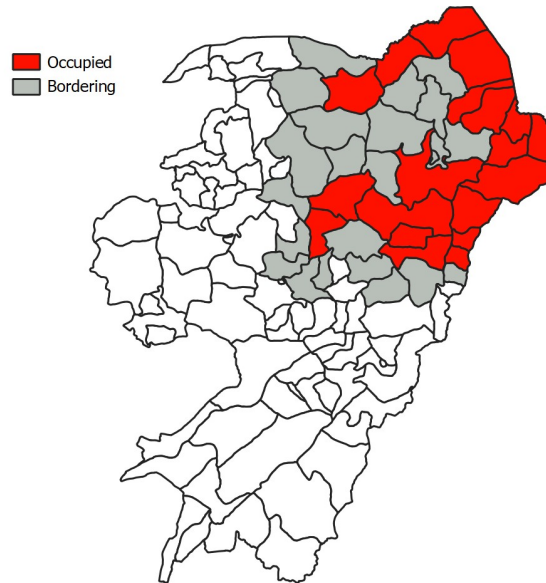
Figure I.2: Data structure



Notes: The diagram explains the data structure. The timeline shows the period considered in this study; the occupation (treatment) took place between 2014 and 2015. Combined, the survey waves provide data on ten school-years. ‘Phone survey’ refers to the phone survey conducted during COVID-19. There are four pre-treatment school-years, after which the treatment (T) took place, followed by six post-treatment time periods.

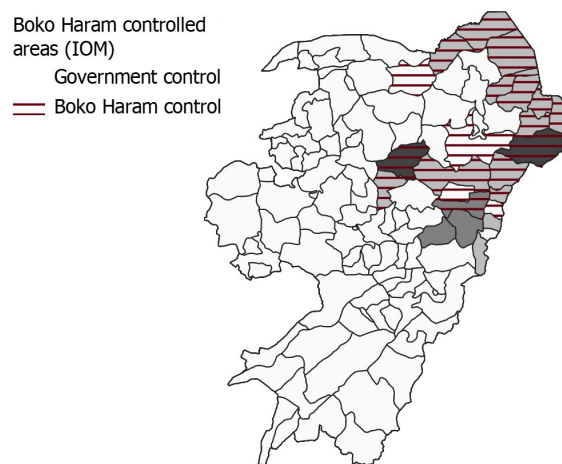
Subsection I.2 Maps

Figure I.3: Occupied and bordering LGAs



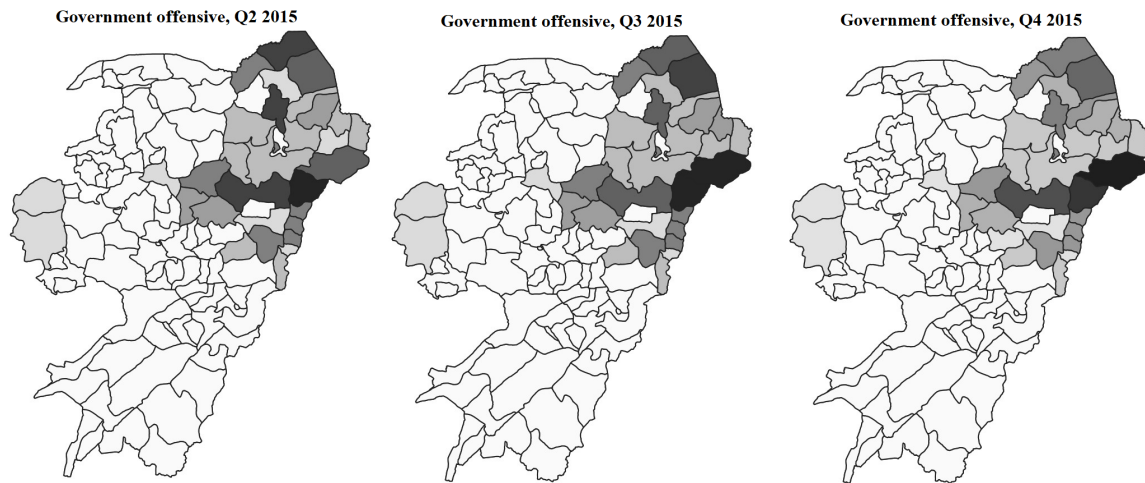
Notes: the map shows the bordering areas (light grey) and the occupied areas (red). Source: IOM and ACLED.

Figure I.4: Occupation of territory by Boko Haram: ACLED and IOM (2015)



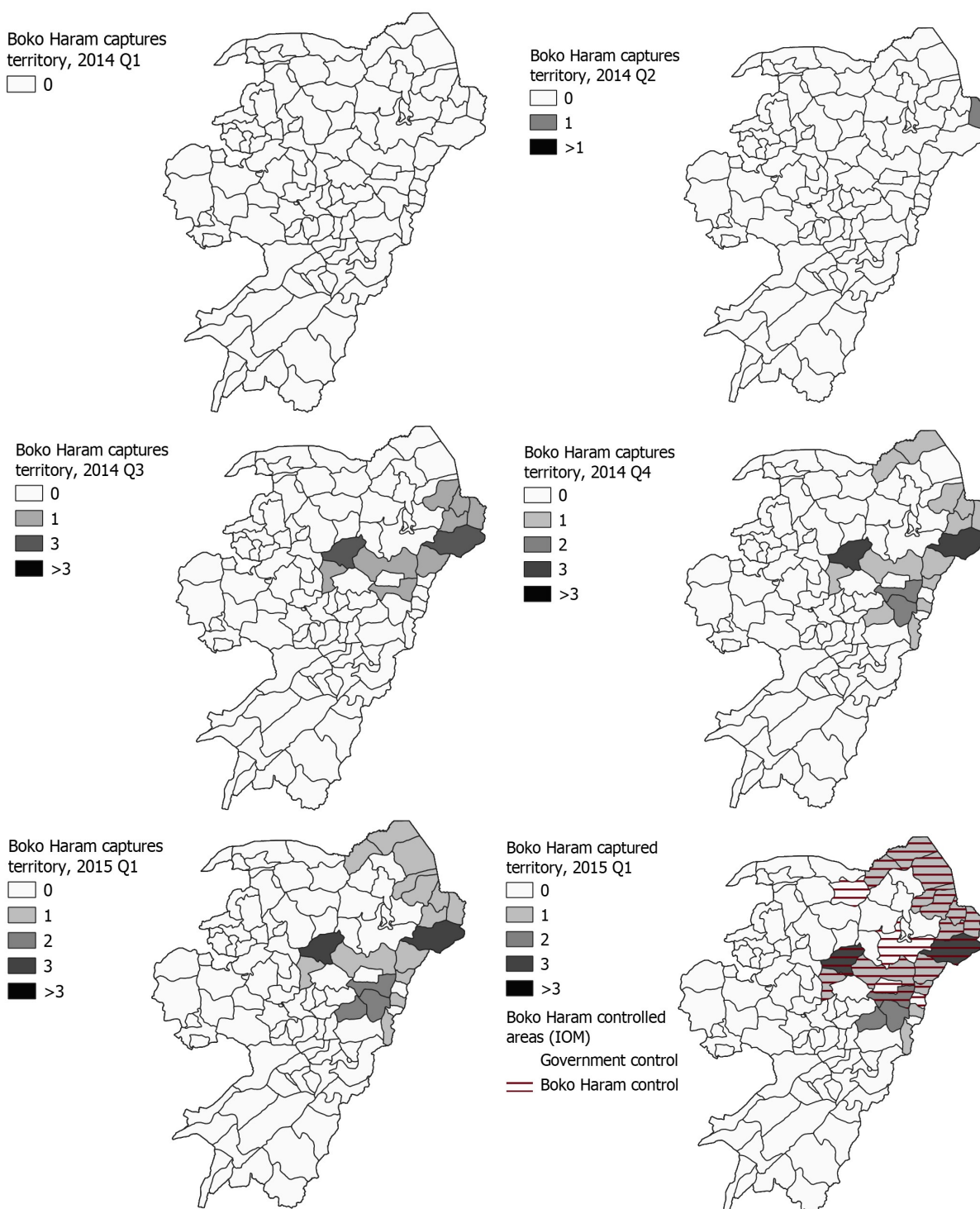
Notes: Maps shows the occupation of territory by Boko Haram in Q1 of 2015, the height of the occupation, as well as the areas that were occupied by the group according to the IOM (striped red). The darker, the more events took place where Boko Haram gained control over territory as recorded by ACLED by quarter/year. Source: ACLED.

Figure I.5: Government offensives to retake territory



Note: Maps shows the progression of the government offensives to retake territory from Boko Haram. The darker, the more events took place where the government regained control over territory as recorded by ACLED by quarter/year. Source: ACLED.

Figure I.6: The progression of the occupation of territory in North-East Nigeria by Boko Haram.



Note: Maps show the number of events, per LGA, where Boko Haram gained control over territory as recorded by ACLED by quarter/year. The last map shows all captured territory as well as the LGAs that were designated by the IOM as being under full Boko Haram control in January 2015. Source: ACLED.